Sahaj Bhadja

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Summary

Software engineer with 2+ years of experience building **mobile-adjacent and full-stack products** across **React/TypeScript, Python APIs, and cloud services**, delivering reliable user-facing workflows, real-time telemetry views, and AI-backed systems in fast-paced robotics-style environments.

Education

University of Southern California **Los Angeles, CA**
Master of Science in Computer Science *August 2024 – May 2026*
GPA: 3.75/4 Coursework: Analysis of Algorithms, Database Systems, Web Technologies

PDEU
B.Tech. in Computer Science and Engineering *Nov 2020 – May 2024*
GPA: 3.88/4 Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Built multi-tenant product features across **FastAPI**, **Azure SQL**, **PostgreSQL**, and cloud storage for 500+ users, reducing dashboard latency from 1.2s to 350ms through schema and query optimization.
- Implemented fault-tolerant background-job and integration workflows for 5K+ monthly tasks with validation and release safeguards, improving throughput 3x while sustaining 99%+ production reliability.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Developed user-facing AI workflows with **Flask** + **React**, secure file ingestion, and modular APIs, handling 10K+ daily requests for 150+ instructors in production.
- Built concurrent processing and release flows on **AWS RDS** with CI/CD-backed deployments, cutting turnaround time by 60% while maintaining stable API response under peak load.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform microservice APIs and data workflows using **Python**, OOP patterns, and **AWS Glue/S3**, improving response times by 30% and pipeline efficiency by 68%.
- Maintained 20+ unit/integration suites with Jenkins, PyTest, and Mockito across distributed environments, strengthening debugging, code review, and release quality in Agile sprints.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a MERN-based operations platform with **REST APIs**, WebSockets, and microservices, supporting 8+ business workflows and 2M+ synchronized records in a user-facing system.
- Implemented modular auth, billing, and notification services with **Jenkins**, **Docker**, and Azure SQL-backed workflows, reducing deployment time by 50% and sustaining 95%+ successful build validation.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built Python-based experimentation and simulation pipelines for autonomous systems, processing 100K+ trajectory logs and improving long-horizon planning performance by 18%.
- Collaborated on model behavior analysis, debugging, and evaluation workflows, improving closed-loop stability by 15% while documenting results for iterative product/research cycles.

Skills

Languages: Python, JavaScript/TypeScript, Java, SQL, C++
Mobile & Frontend: React, React Native, Flutter, HTML/CSS, TailwindCSS, Responsive UI, App Store/Play Store Release Concepts
Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST APIs, JSON, OOP, WebSocket Integration
Databases: PostgreSQL, MySQL, MongoDB, SQL Server, Azure SQL, Relational Data Modeling
Cloud & Infrastructure: AWS (RDS, S3, Glue), Microsoft Azure, Docker, Kubernetes, Jenkins, Git/GitHub
AI/ML & Robotics: LLM Systems, Computer Vision Workflows, Intelligent Automation, Prompt Engineering
Engineering Practices: CI/CD, Unit Testing, Integration Testing, Agile/Scrum, Production Debugging, Reliability Monitoring

Projects

PiXMod | *Next.js, TailwindCSS, MongoDB, Stripe, Transformers*
- Built an AI-powered SaaS application with responsive user experiences, image-processing workflows, and scalable storage in MongoDB, increasing premium conversion by 18%.

FinNexus | *Next.js, TypeScript, Kubernetes, MongoDB*
- Built a full-stack fintech product with real-time updates and secure transaction controls, supporting 50K+ users and 1M+ transactions across a distributed deployment stack.

CPU-Scheduling Simulator | *C++, React.js, JavaScript*
- Implemented 6 scheduling algorithms with interactive controls and live visualizations across 1K+ simulated process combinations, strengthening systems reasoning and debugging workflows.